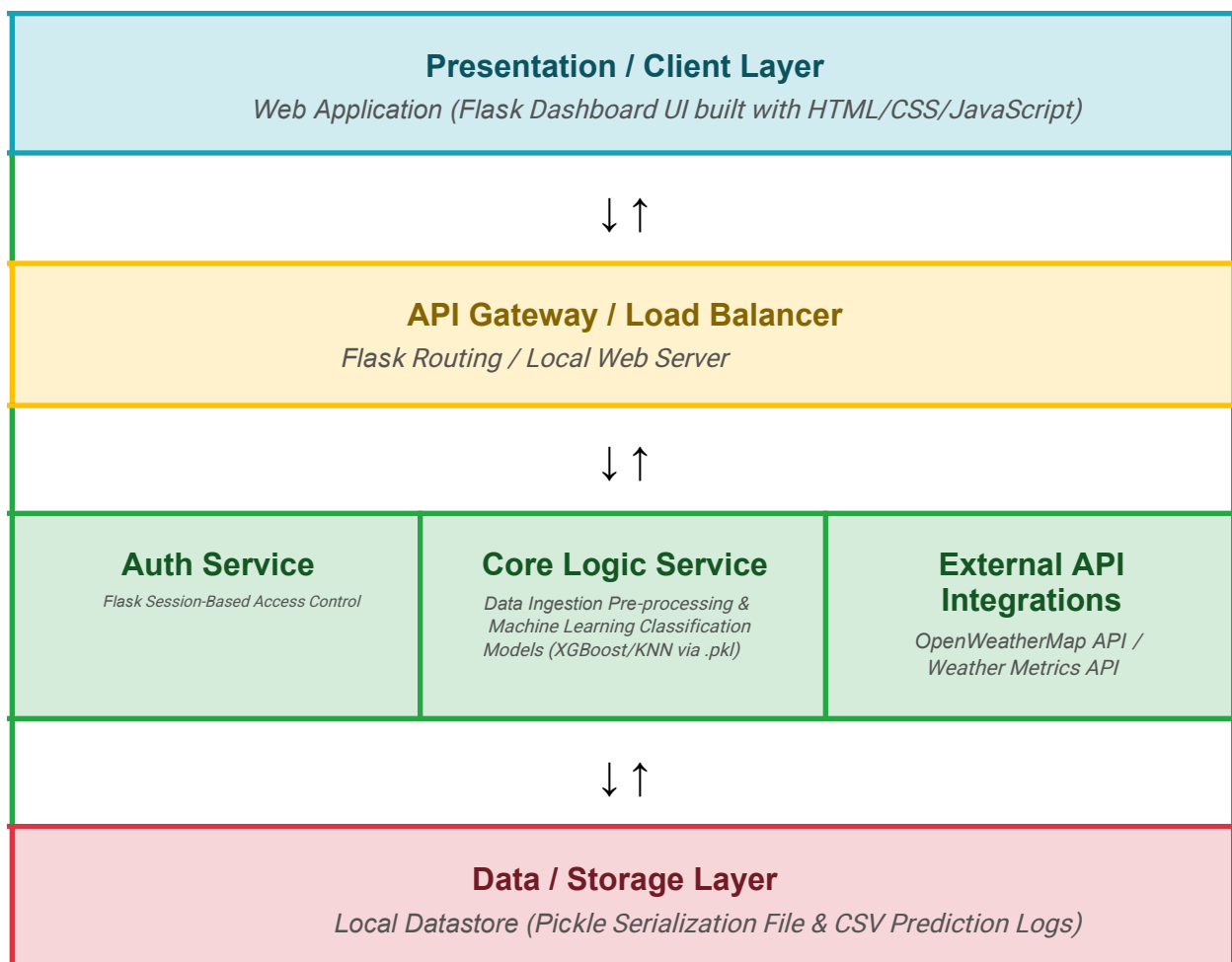


Solution Architecture

Date	10 July 2026
Team ID	XXXXXX
Project Name	Raising Waters
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>[Presentation Layer]</i>	<i>Renders the web interface for the Disaster Coordinator to input monitoring parameters, visually observe early-warning alerts, and scan the real-time flood risk tier dashboard.</i>	<i>HTML5, CSS3, JavaScript</i>
<i>[API Gateway]</i>	<i>Handles inbound routing requests from the presentation panel and safely channels structured requests directly to backend inference paths.</i>	<i>Flask Native Routing Layer</i>
<i>[Microservices]</i>	<i>Core Logic Service: Ingests live environmental streams, transforms metrics, and executes predictive calculations through trained model weights to output risk tiers</i>	<i>Python, Scikit-learn, OpenWeatherMap API</i>
<i>[Database]</i>	<i>External Integrations: Automatically requests live meteorological tracking data from a remote cloud provider</i>	<i>File-system Storage (.pkl, Local CSV)</i>